

The K-theoretic DT/PT vertex correspondence

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September 11, 2023

Joint work with Nick Kuhn & Felix Thimm

[arXiv: 23?? . ?????]

Donaldson–Thomas theory

Setting:

X = smooth (quasi-)projective 3-fold over $\mathbb{C}$

$\mathcal{M}^{\text{DT}}(X) = \{\text{ideal sheaves of curves on } X\}.$

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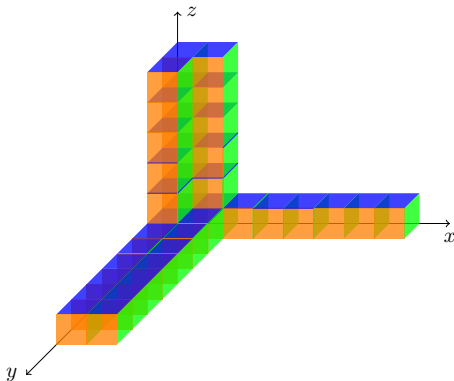
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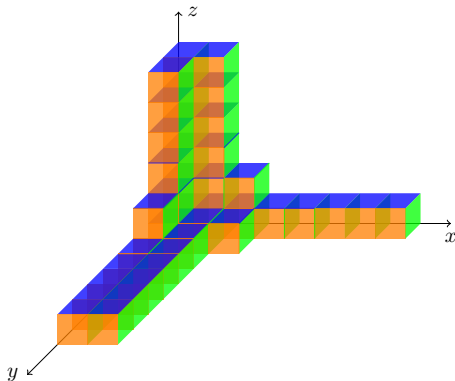
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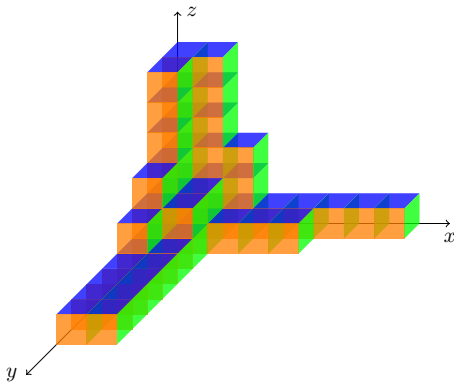
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E.g. $V_{\emptyset, \emptyset, \emptyset}^{\text{DT}, K}(\text{non-equivariant}) = \prod_{n>0} (1 - Q^n)^{-n}$ is MacMahon's famous enumeration of 3d partitions.

Pandharipande–Thomas theory

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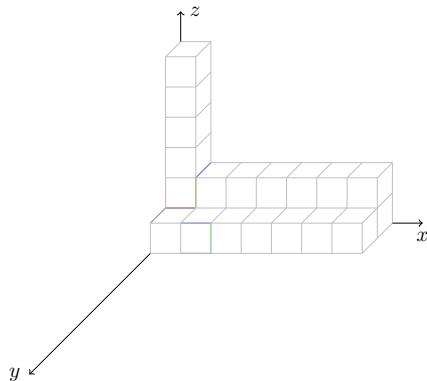
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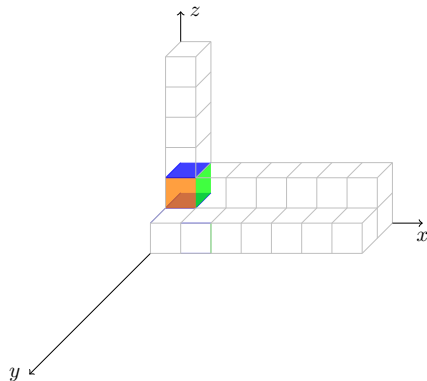
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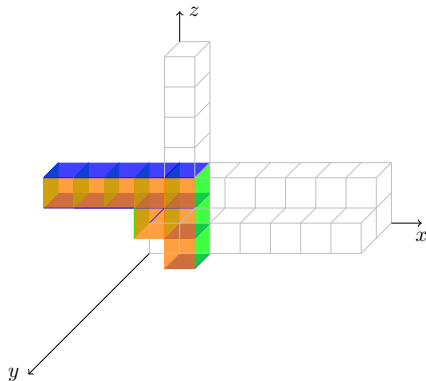
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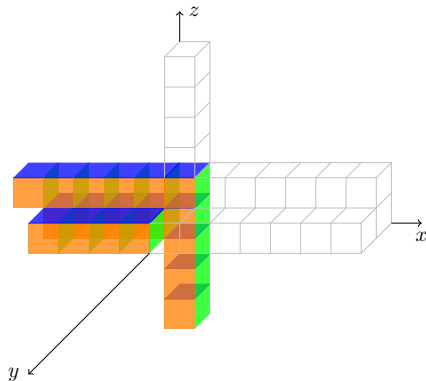
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$$V_{\emptyset, \emptyset, \emptyset}^{\text{PT}, K}(x, y, z) = 1.$$

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Conjecture (PT '07)

$$V_{\lambda, \mu, \nu}^{\text{DT}} = V_{\lambda, \mu, \nu}^{\text{PT}} V_{\emptyset, \emptyset, \emptyset}^{\text{DT}}.$$

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Indeed, we use wall-crossing techniques (which work equally well in equivariant cohomology).

DT/PT (vertex) correspondence

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$$Z_{X,\beta}^M := \sum_{n \in \mathbb{Z}} Q^n \int_{[\mathcal{M}_{\beta_C,n}^M(X)]^{\text{vir}}} 1, \quad M \in \{\text{DT}, \text{PT}\}.$$

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Previous work also includes combinatorial approaches to the numerical DT/PT vertex correspondence

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(Jenne–Webb–Young '20) Holds for arbitrary λ, μ, ν . (Hard!)

DT/PT vertex correspondence

The full $V_{\lambda,\mu,\nu}^{M,K}(x,y,z)$ are **genuinely equivariant** objects, much more sophisticated than $Z_{X,\beta}^{M,K}$, $Z_{X,\beta}^M$, or $\lim_{xyz \rightarrow 1} V_{\lambda,\mu,\nu}^{M,K}$.

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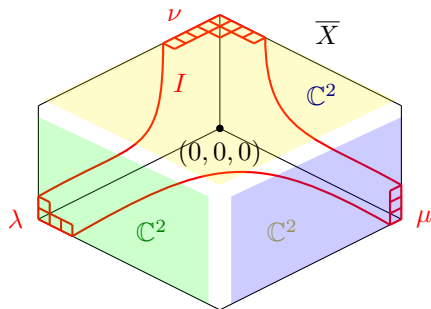
We always work equivariantly with respect to the obvious action of $T = (\mathbb{C}^\times)^3$.

Proof strategy, step 1: formulate the wall-crossing problem

Let $\overline{X} := \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ and fix λ, μ, ν .

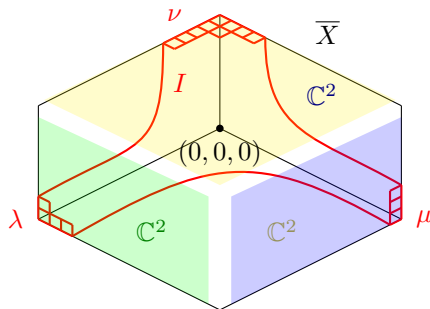
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There is a **moduli stack** $\mathfrak{N}_{(\lambda, \mu, \nu), n}$ of pairs $I = [\mathcal{O}_{\overline{X}} \rightarrow \mathcal{E}]$ with this prescribed (derived) restriction to the boundary $D = D_1 \cup D_2 \cup D_3$, and **numerical class**

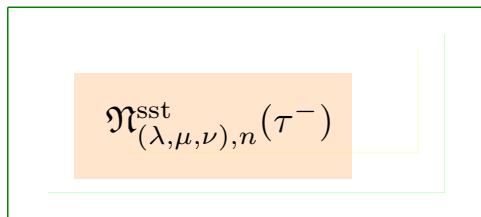
$$\text{ch}(I) = (1, 0, -\beta_C, -n), \quad \beta_C := (|\lambda|, |\mu|, |\nu|).$$

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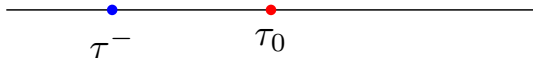
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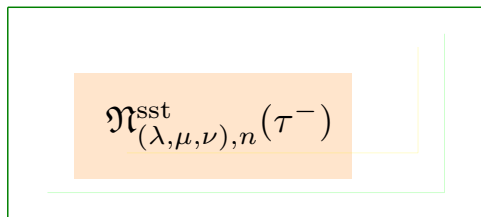

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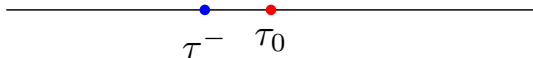


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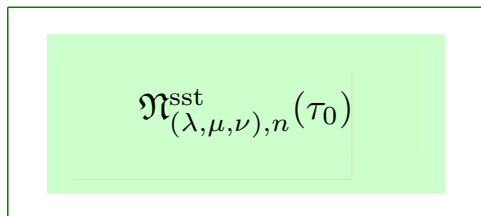

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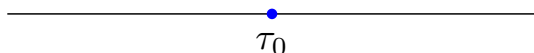


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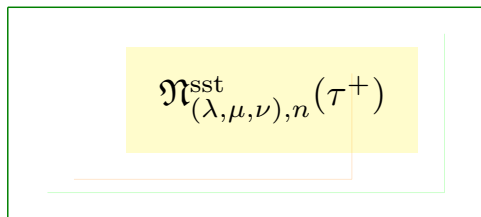

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$$[\mathcal{O} \rightarrow \mathcal{E}] \oplus [0 \rightarrow \mathcal{Q}_1] \oplus \cdots \oplus [0 \rightarrow \mathcal{Q}_k]$$

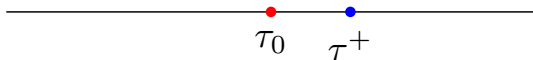

$$\tau_0$$

Proof strategy, step 1: formulate the wall-crossing problem

Toda has a continuous family $(\tau_\xi)_\xi$ of **weak stability conditions**, on the underlying abelian category $\langle \mathcal{O}_{\overline{X}}, \text{Coh}(\overline{X})[-1] \rangle$, such that:


$$\mathfrak{N}_{(\lambda, \mu, \nu), n}^{\text{sst}}(\tau^+)$$

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Proposition

There is a *symmetric* obstruction theory on $\mathfrak{N}_{(\lambda,\mu,\nu),n}$, given by $\mathrm{Ext}_{\overline{X}}(I, I(-D))$, such that

$$V_{\lambda,\mu,\nu}^{\mathrm{DT},K} = \sum_n Q^N \chi(\mathfrak{N}_{(\lambda,\mu,\nu),n}^{\mathrm{sst}}(\tau^-), \hat{\mathcal{O}}^{\mathrm{vir}})$$
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Nekrasov–Okounkov *symmetrization* $\hat{\mathcal{O}}^{\mathrm{vir}} := \mathcal{K}_{\mathrm{vir}}^{1/2} \otimes \mathcal{O}^{\mathrm{vir}}$.

Proof strategy, step 2: add framing data to the invariants

Problem: a stable object $I = [\mathcal{O}_X \rightarrow \mathcal{E}]$ can **split** into a sum of $N \geq 1$ **strictly-semistable pieces** at the wall τ_0 . But **master space** arguments (later) require $N \leq 2$.

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$$\begin{array}{ccccccc} V_1 & \xrightarrow{\rho_1} & V_2 & \xrightarrow{\rho_2} & \dots & \xrightarrow{\rho_{N-1}} & V_N & \xrightarrow{\rho_N} & V_{N+1} = F_{k,p}(I) \\ \blacksquare & & \blacksquare & & & & \blacksquare & & \bullet \end{array}$$

for $\dim \mathbf{V} = \mathbf{d}$ and the **framing functor**

$$F_k([L \otimes \mathcal{O}_X \rightarrow \mathcal{E}]) := H^0(\mathcal{E}(k)) \oplus L.$$

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Lift $\tau^\pm$ from $\mathfrak{N}$ to (weak) stability conditions $\tilde{\tau}^\pm$ on $\mathfrak{N}^{Q(N)}$.

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- have all ρ_i injective;

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Forgetful map $\Pi_{\mathfrak{N}}: (I, \mathbf{V}, \rho) \mapsto I$ has fibers = full flag varieties.

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$M_{\alpha_1}^{\text{sst}}(\tilde{\tau}_0) \times M_{\alpha_2}^{\text{sst}}(\tilde{\tau}_0)$ for (special) splittings $\alpha = \alpha_1 + \alpha_2$.

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$$\chi(\mathbb{M}, \mathcal{O}^{\text{vir}}) = \sum_{F \subset \mathbb{M}^{\text{C}\times}} \chi \left(F, \frac{\mathcal{O}_F^{\text{vir}}}{\wedge_{-1}^{\bullet} (\mathcal{N}^{\text{vir}})^{\vee}} \right)$$

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where $\mathbb{k} := \mathbb{Z}[x^{\pm}, y^{\pm}, z^{\pm}, u^{\pm 1/2}]$.

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(from properness of $\mathbb{M}^T$ + a bit more)

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where $\mathbb{k} := \mathbb{Z}[x^\pm, y^\pm, z^\pm, u^{\pm 1/2}]$. Apply **K-theoretic residue map**:

$$0 = \sum_{F \subset \mathbb{M}^{\mathbb{C}^\times}} \text{Res}_u^K \chi \left(F, \frac{\hat{\mathcal{O}}_F^{\text{vir}}}{\hat{\wedge}_{-1}^\bullet(\mathcal{N}^{\text{vir}})^\vee} \right).$$

where $\text{Res}_u^K(f) := (\text{res}_{u=0} + \text{res}_{u=\infty})(f \, u \, du)$.

Proof strategy, step 3: construct master spaces

$$0 = \sum_{F \subset \mathbb{M}^{\times}} \operatorname{Res}_u^K \chi \left(F, \frac{\hat{\mathcal{O}}_F^{\text{vir}}}{\hat{\wedge}_{-1}^{\bullet} (\mathcal{N}^{\text{vir}})^{\vee}} \right).$$

The goal: put an obstruction theory on $\mathbb{M}$ such that these residues are **understandable** and **explicit**.

Interlude: symmetric obstruction theories

A symmetric (perfect) obstruction theory $\mathbb{E} \in D_{\mathrm{QCoh}}$ satisfies

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for $\kappa := xyz$.

(Restriction to any semistable = stable locus is **automatically perfect!**)

Interlude: symmetric obstruction theories

Key observation: if $\mathcal{N}^{\text{vir}} = \mathcal{F} - \kappa^{-1} \otimes \mathcal{F}^\vee$ is symmetric, then

$$\frac{1}{\widehat{\Lambda}_{-1}^\bullet(\mathcal{N}^{\text{vir}})^\vee} = \prod_{w \in \mathcal{F}} -\frac{(\kappa w)^{1/2} - (\kappa w)^{-1/2}}{w^{1/2} - w^{-1/2}}.$$

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$$\text{Res}_u^K \frac{1}{\widehat{\wedge}_{-1}^\bullet(\mathcal{N}^{\text{vir}})^\vee} = (-1)^{\text{ind}} (\kappa^{\text{ind}/2} - \kappa^{-\text{ind}/2})$$

where $\text{ind} := \text{rank } \mathcal{F}_{>0} - \text{rank } \mathcal{F}_{<0}$ is a kind of [Morse–Bott index](#) of each $\mathbb{C}^\times$ -fixed component.

Interlude: symmetric obstruction theories

Our situation: **smooth** morphisms of Artin stacks

$$\mathbb{M} \quad \text{“}\rightarrow\text{”} \quad \mathfrak{N}_{(\lambda, \mu, \nu), n, \mathbf{d}}^{Q(N)} \quad \rightarrow \quad \mathfrak{N}_{(\lambda, \mu, \nu), n}.$$

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Technical heart of our work: **symmetrized pullback** of symmetric obstruction theories.

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Let $f: \mathfrak{X} \rightarrow \mathfrak{Y}$ be a **smooth** morphism of Artin stacks, and $\phi: \mathbb{E}_{\mathfrak{Y}} \rightarrow \mathbb{L}_{\mathfrak{Y}}$ be a **symmetric** obstruction theory.

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$$\begin{array}{ccccccc} & & f^* \mathbb{E}_{\mathfrak{Y}} & & & & \\ & & \downarrow f^* \phi & & & & \\ \mathbb{L}_f[-1] & \longrightarrow & f^* \mathbb{L}_{\mathfrak{Y}} & \longrightarrow & \mathbb{L}_{\mathfrak{X}} & \xrightarrow{+1} & \end{array}$$

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(a *non-symmetric* obstruction theory for $\mathfrak{X}$)

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the desired (symmetric?) obstruction theory for $\mathfrak{X}$

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None of these hold in general, but they do hold on any **affine chart** for degree reasons.

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Theorem (Kiem–Savvas '20, '21)

*APOTs induce virtual structure sheaves $\mathcal{O}_{\mathfrak{X}}^{\text{vir}}$. They satisfy equivariant localization **assuming $\mathcal{N}^{\text{vir}}$ also exists globally.***

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Pieces of DT and PT vertices respectively.

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The κ -symmetry of (AP)OTs **quantizes** many integers n into the **(symmetric) quantum integers**

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Similarly, integration along full flag varieties produces $[N]_{\kappa}! := [N]_{\kappa} [N-1]_{\kappa} \cdots [1]_{\kappa}$.

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(Can quantize many formulas in Joyce–Song this way.)

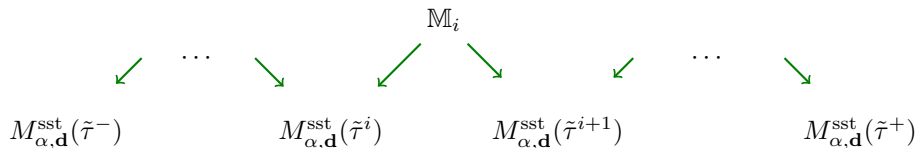
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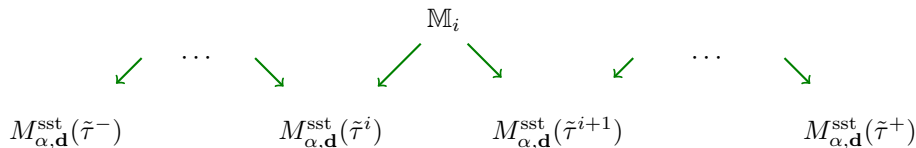
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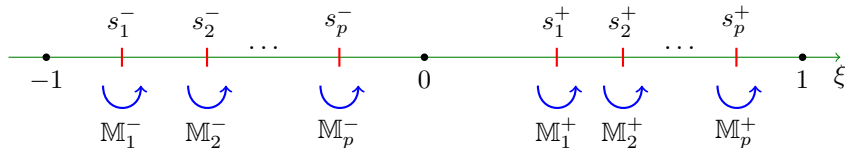
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Joyce-style (“universal wall-crossing”): for $M_{\alpha, \mathbf{d}}^{\text{sst}}(\tilde{\tau}_\xi)$,



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For both, the “single-step” wall-crossing formula has the form

$$\begin{aligned} 0 = & \tilde{N}_{(\lambda, \mu, \nu), n, \mathbf{d}}(s_+) - \tilde{N}_{(\lambda, \mu, \nu), n, \mathbf{d}}(s_-) \\ & + \sum_{\substack{m \in \mathbb{Z}_{>0} \\ \mathbf{d} = \mathbf{e} + \mathbf{f} \\ \dots}} [m - c(\mathbf{e}, \mathbf{f})]_{\kappa} \cdot \tilde{Q}_{m, \mathbf{e}} \tilde{N}_{(\lambda, \mu, \nu), n-m, \mathbf{f}} \end{aligned}$$

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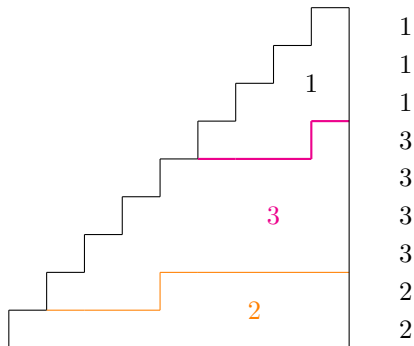
Iterate to get the full wall-crossing formula.

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Iterated splittings of a full flag $\mathbf{d} = (1, 2, \dots, N)$ into smaller full flags are equivalent to **word rearrangements**:

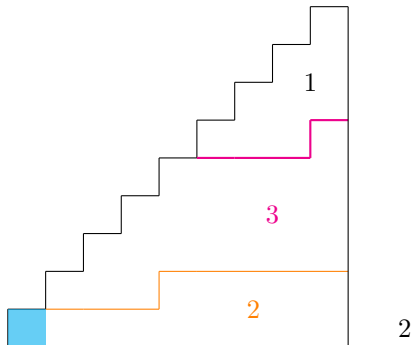
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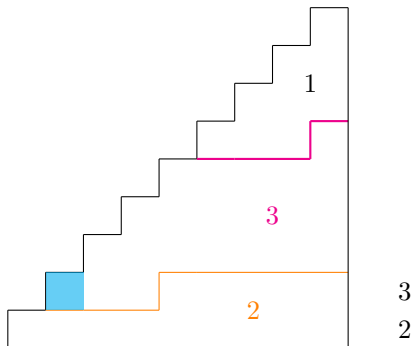
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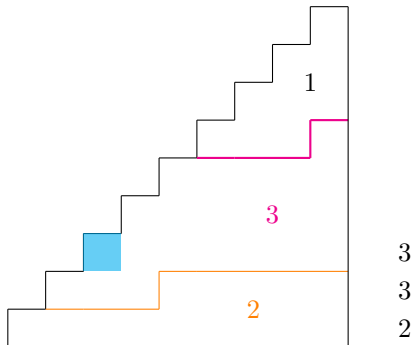
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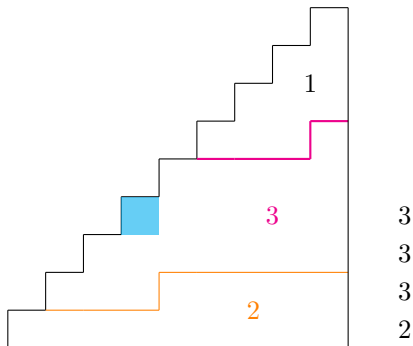
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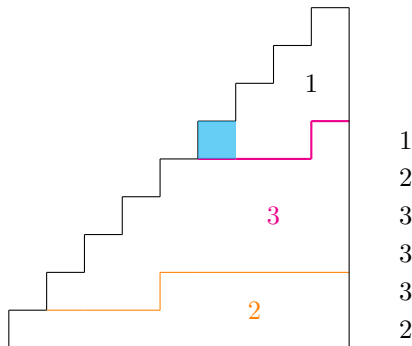
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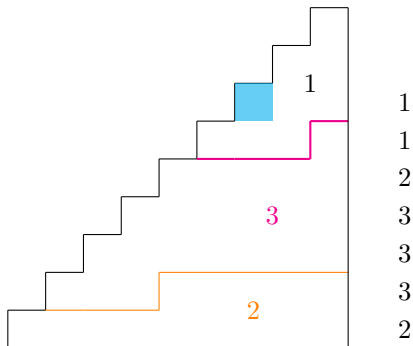
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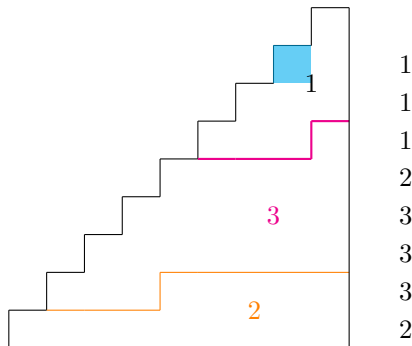
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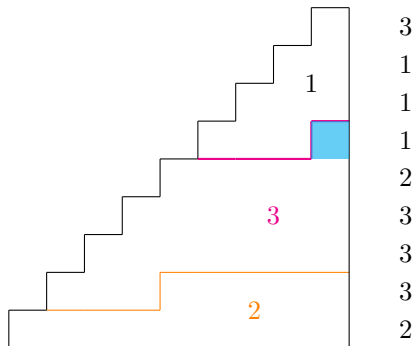
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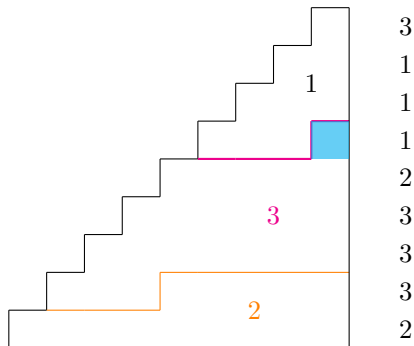
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$$R(m_1, \dots, m_k) := \{\text{rearrangements of } 1^{m_1} 2^{m_2} \dots k^{m_k}\}.$$

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Proposition

For $k \geq 0$ and $m_1, \dots, m_k, m_{k+1} \geq 1$,

$$\frac{1}{k!} \sum_{\sigma \in S_k} \sum_{\substack{w \in R(m_1, \dots, m_{k+1}) \\ o_{\sigma(1)}(w) > \dots > o_{\sigma(k)}(w)}} \prod_{i=1}^k [m_{\sigma(i)} - \sum_{j=i+1}^{k+1} c(\mathbf{e}_{\sigma(i)}, \mathbf{e}_{\sigma(j)})]_{\kappa} = \frac{[m_1 + \dots + m_{k+1}]_{\kappa}!}{[m_{k+1}]_{\kappa}! \prod_{i=1}^{k-1} [m_i - 1]_{\kappa}!}$$

where $o_i(w)$ is the index of the first occurrence of i in w , and $c(\mathbf{e}_i, \mathbf{e}_j) \approx$ the number of **inversions** in w for i and j .

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word combinatorics is much harder, but can be sidestepped by a trick using the freedom to choose $p \geq 1$ in the framing functor $F_{k,p} = \cdots \oplus L^{\oplus p}$.

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Finally, the Joyce-style approach may also provide formulas for DT/PT *descendent transformations*.

Thank you!

